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OUTCOMES-BASED COURSE SYLLABUS

College/Institute	**COLLEGE OF ENGINEERING**
Department	**CHEMAPHY**
Program	**ALL ENGINEERING PROGRAMS**

RTU Vision	
A SMART and transnational university. *Isang SMART at transnasyonal na unibersidad.*	
RTU Mission	**RTU Philosophy**
To produce competent and socially responsible professionals through innovative instructions, high-impact research, and sustainable extension services responsive to the diverse needs of glocal communities. *Makalikha ng mga propesyonal na may kakayahan at may gampaning panlipunan sa pamamagitan ng inobatibong pagtuturo, mataas na antas ng pananaliksik, at nagpapatuloy na programang pang-ekstensiyon na tumutugon sa iba't ibang pangangailangan ng mga glokal na komunidad.*	**The Rizal Technological University believes in nurturing the creative potentials of Filipinos to excel in a dynamic world order and advocates commitment to global peace and sustainable development along with sense of moral responsibility and cultural patronage.** *Naniniwala ang Rizal Technological University sa pangangalaga sa malikhaing potensyal ng mga Filipino upang manguna sa pabago-bagong takbo ng mundo at magtaguyod ng komitment patungo sa pandaigdigang kapayapaan at napananatiling pag-unlad kasabay ng moral na responsibilidad at kultural na pagtangkilik.*
Core Values	
Integrity, Nationalism, Inclusivity, Excellence, Innovation *Integridad, Nasyonalismo, Pagka-inklusibo, Kahusayan, Inobasyon*	

College/Institute Goal	Provide opportunities for qualified student to become professional engineers and technicians of the country.
College/Institute Objectives	• Acquire full understanding of the scientific principle and knowledge of the particular field; • Develop a high level of competence in its engineering methods and applications; • Communicate effectively and succinctly important result of any technical study both verbally and technically; • Nurture the desire of continuing professional growth and explore new horizons in technology; and, • Imbue the graduates with socially and morally sound motivations and principles to live by.

Program Educational Objectives		
Program Outcomes with Performance Indicators	**POs***	**PIs**
	1. Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.	**Problem Decomposition and Principles Identification:** The student correctly identifies and articulates the fundamental mathematical, scientific, and engineering principles underlying a complex engineering problem, breaking it down into its key components. **Appropriate Model Selection and Application:** The student selects and applies appropriate theoretical concepts, mathematical models, and computational tools from both foundational and specialized engineering knowledge to analyze the defined problem. **Validated and Justified Solution:** The student formulates a viable solution path and arrives at a conclusion that is technically sound, properly validated against established principles, and clearly justified based on the applied analysis.
	2. Conduct investigations of complex engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	**Research and Experimental Design:** The student designs and documents a systematic investigation or experiment, grounded in research-based knowledge, to address a complex problem, including a justified methodology and variables. **Data Analysis and Interpretation:** The student competently collects, processes, and analyzes experimental or research data using appropriate quantitative or qualitative techniques, identifying key trends, patterns, and anomalies. **Evidence-Based Conclusion:** The student synthesizes the results of the investigation to draw valid, well-reasoned conclusions that are explicitly supported by the interpreted data and address the original problem.
	3. Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration	**Requirement Synthesis:** The student generates a design specification that accurately translates the defined needs and constraints, including explicit considerations for public health, safety, and environmental impact.

	for public health and safety, cultural, societal, and environmental considerations.	**Conceptual Design and Evaluation:** The student creates multiple feasible design concepts and conducts a comparative analysis, justifying the selected approach based on how it balances performance with societal, cultural, and environmental factors. **Validated Final Design:** The student produces a detailed final design with appropriate documentation, demonstrating through analysis or reference to standards that the solution meets the specified needs while mitigating identified societal and safety risks.
	4. Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	**Individual Contribution and Accountability:** The student demonstrates personal responsibility by consistently completing assigned tasks on time and meeting quality standards, thereby fulfilling their role within the team. **Collaborative Integration:** The student actively listens, provides constructive feedback, and integrates the ideas and expertise of other members to advance the team's collective work on a project. **Leadership and Conflict Management:** The student, when acting as a member or leader, facilitates group discussions, helps manage project timelines, and employs effective strategies to resolve interpersonal conflicts constructively.
	5. Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.	**Problem Scoping and Formulation:** The student accurately identifies the core elements of a complex, ill-defined problem and formulates a precise, researchable engineering problem statement. **Research and Principle Integration:** The student conducts a systematic literature review to contextualize the problem and correctly identifies and applies relevant first principles from mathematics, natural sciences, and engineering sciences to analyze it. **Substantiated Conclusion:** The student synthesizes their analysis and research findings to reach a logical, well-reasoned conclusion that is explicitly substantiated by the evidence and principles applied.

	6. Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.	**Ethical Dilemma Identification:** The student can identify and articulate the ethical dimensions, professional responsibilities, and potential conflicts of interest within a given engineering scenario. **Principles Application:** The student analyzes an ethical dilemma by applying relevant sections from recognized engineering codes of ethics (e.g., IEEE, NSPE) to propose a course of action. **Commitment to Public Welfare:** The student consistently prioritizes public health, safety, and well-being in design decisions and project justifications, explaining the societal impact of their work.
	7. Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	**Technical Documentation:** The student produces clear, well-structured, and comprehensive technical reports or design documentation that effectively convey complex information to the engineering community. **Adapted Oral Presentation:** The student delivers a compelling oral presentation, tailoring technical content and language appropriately for the audience, whether peers or society at large. **Active Listening and Instruction:** The student demonstrates comprehension of complex instructions and technical discussions by accurately summarizing key points and providing relevant, clarifying questions or feedback
	8. Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental context.	**Impact Assessment:** The student identifies and analyzes both the potential long-term societal benefits and the environmental consequences of a proposed engineering solution. **Sustainable Design Evaluation:** The student evaluates alternative solutions or design choices using sustainability criteria, such as resource efficiency, lifecycle analysis, and environmental impact. **Contextual Justification:** The student justifies engineering decisions by explicitly discussing their alignment with

		sustainable development goals and their broader impact on community well-being and ecological systems.
	9. Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	**Self-Directed Learning Initiative:** The student independently identifies a knowledge gap related to technological change and seeks out relevant learning resources to address it. **Skill Adaptation:** The student demonstrates the ability to learn and apply a new tool, technique, or software platform to complete an engineering task, showing adaptation to evolving technologies. **Reflective Practice:** The student articulates a plan for their continued professional development, reflecting on past learning experiences to guide future skill acquisition in a rapidly changing field.
	10. Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.	**Contextual Issue Identification:** The student systematically identifies and describes the broader societal, health, safety, legal, and cultural issues relevant to a given engineering problem or proposed solution. **Informed Risk and Impact Assessment:** The student uses reasoned judgment, supported by research and contextual knowledge, to assess the potential risks, impacts, and consequent responsibilities arising from professional engineering decisions. **Responsibility-Mitigation Integration:** The student proposes specific, actionable measures within an engineering solution or professional practice to mitigate identified negative impacts and fulfill relevant professional and societal responsibilities.
	11. Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems with an understanding of the limitations.	**Tool Selection and Justification:** The student selects and justifies the use of specific modern engineering tools, software, or IT resources as being appropriate for modeling or solving a given complex problem.

		Competent Application and Creation: The student competently applies the selected tools to create a model, run a simulation, or analyze data, demonstrating proficiency in their operational use. **Critical Evaluation of Limitations:** The student critically discusses the assumptions, uncertainties, and limitations of the tools and techniques used, and contextualizes the results accordingly.
	12. Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	**Project Scoping and Planning:** The student develops a realistic project plan that includes a timeline, resource allocation, and a budget, applying fundamental engineering management principles. **Economic Analysis:** The student performs and justifies an economic analysis (e.g., cost-benefit, ROI) to support technical decisions and evaluate project feasibility. **Team Management and Execution:** The student effectively manages a team task by delegating work, monitoring progress against the plan, and making adjustments to mitigate risks and ensure timely completion.

COURSE INFORMATION						
Course Code	**GE04**	**Course Title**	**Mathematics in the Modern World**		**Credit Units**	3
Pre-requisite	none		**Co-requisite**	none	**Contact Hours**	3 Hours Lecture/Week
Course Description	This course deals with nature of Mathematics, appreciation of its practical, intellectual, and aesthetic dimension and application of mathematical tools in daily life. The course begins with an introduction to the nature of mathematics as an exploration of patterns (in nature and the environment) and as application of inductive and deductive reasoning. By exploring these topics, students are encouraged to go beyond the typical understanding of mathematics as merely a set of formulas but as a source of aesthetics in patterns of nature for example and a rich language in itself (and of science) governed by logic and reasoning. The course then proceeds to survey ways in which mathematics provides a tool for understanding and dealing with various aspects of present day living such as managing personal finances, making social choices, appreciating geometric designs, understanding codes used in data					

	transmission and security and diving limited resources daily. These aspects will provide opportunities for actually doing mathematics in a broad range of exercises that bring out the various dimensions of mathematics as a way of knowing and test the students' understanding and capacity (CMO No. 20, series of 2013)					
Semester		**Academic Year**		**Instructor/Professor**		
Course Outcomes *Legend: **I** - Introduced (an introductory course to an outcome); **E** – Enabled (a course that strengthens the outcome); **D** – Demonstrated (a course demonstrating an outcome)*	**PO1**	**PO4**	**PO5**	**…**	**PO8**	
Discuss and argue about the nature of mathematics, what it is, how it is expressed, represented and used.	I	I	I		I	
Use different types of reasoning to justify statements and arguments made about mathematics and mathematical concepts	I	I	I		I	
Discuss the language and symbols of mathematics.	I	I	I		I	
Use a variety of statistical tools to process and manage numerical data.	I	I	I		I	
Analyze codes and coding schemes used for identification, privacy and security.	I	I	I		I	
Use mathematics in other areas such as finance, voting, health and medicine, business, environment, arts and design, and recreation.	I	I	I		I	
Appreciate the nature and uses of mathematics in everyday life.	I	I	I		I	
Affirm honesty and integrity in the application of mathematics to various human endeavors.	I	I	I		I	

TEACHING AND LEARNING PLAN

Source: Handbook on Typology, Outcomes-Based Education, and Institutional Sustainability Assessment; CHED PSGs

Session	Learning Outcomes	Topics	Teaching-Learning Activities	Teaching-Learning Resources	Assessment Tasks/Tools	Evaluation Methods	Output/ Requirements
1-4	**CO1, CO7 & CO8** **ORIENTATION**	**Chapter 1: The Nature of Mathematics** A. Patterns and Numbers in Nature and the World B. The Fibonacci Sequence	1. Lecture through PowerPoint presentation 2. Video Watching		Essay writing Synthesis Paper Quiz	1. Written Essay 2. Synthesis Paper 3. Activities/exercises 4. Quiz	1. Activities/ Exercises

	LO1: Identify patterns in nature and regularities in the world. LO2: Articulate the importance of mathematics in one's life. LO3: Argue about the nature of mathematics what it is, how it is expressed, represented and used. LO4: Express appreciation for mathematics as a human endeavor.	C. Patterns and regularities in the world as organized by Mathematics D. Phenomena in the world as predicted by Mathematics E. Nature and occurrences in the world as controlled by Mathematics for human ends Applications of Mathematics in the world	3. Send learning materials via LMS/group chat	Workbook, reference books			2. Quiz **Target: 60%**
5-8	**CO1 & CO3** LO1: Discuss the language, symbol, and conversation of mathematics. LO2: Explain the nature of mathematics as and language. LO3: Perform operations on mathematical expressions correctly, and, LO4: Acknowledge that mathematics is a useful language	**Chapter 2: Mathematical Language and Symbols** A. Characteristics of mathematical language; precise, concise, powerful B. Expressions vs. sentences C. Conventions in the mathematical language D. Four basic concepts: sets, functions, relations, binary operations E. Elementary logic; connectives, quantifiers, negation, variables Formality	• Problem Solving • Interactive Learning • Brainstorming • Multimedia/Videos • Synchronous Discussion	Workbook, reference books	Quizzes and problem sets	1.Questionnaires 2. Use of scoring rubrics	1. Activities/ Exercises 2. Quiz **Target: 60%**
9	**PRELIMINARY EXAMINATION** **Target: 60%**						

10-16	**CO2 & CO3** LO1: Use different types of reasoning to justify statements and arguments made about mathematics and mathematical concepts LO2: Write clear and logical proofs. LO3: Solve problems involving patterns and recreational problems following Polya's four steps LO4: Organize one's method and approaches for proving and solving problems.	**Chapter 3: Problem Solving and Reasoning** A. Inductive and Deductive Reasoning B. Intuition, proof, and certainty C. Polya's 4-steps in Problem Solving D. Problem solving strategies E. Mathematical Problems involving Patterns 1. Recreational Problems using mathematics	1.Lecture through PowerPoint presentation. 2. Send learning materials in our LMS/Group Chat 3. Ask them to watch a recorded video lecture via YouTube channel	Workbook, reference books	Quizzes and problem sets	1. Questionnaires 2. Use of scoring rubrics	1. Activities/ Exercises 2. Quiz **Target: 60%**
17-22	**CO4** LO1: Use a variety of statistical tools to process and manage numerical data LO2: Use the methods of linear regression and correlations to predict the value of a variable given certain condition. LO3: Advocate the use of statistical data in making important decisions.	**Chapter 4: Statistics** A. Gathering, Organizing, Representing, Analyzing and Interpreting Data B. Measures of Central Tendency C. Measures of Dispersion D. Measures of Relative Position E. Probabilities and Normal Distribution Correlation ang Regression	1.Lecture through PowerPoint presentation. 2. Send learning materials in our LMS/Group Chat 3. Ask them to watch a recorded video lecture via YouTube channel	Workbook, reference books	Quizzes and problem sets	1. Questionnaires 2. Use of scoring rubrics	1. Activities/ Exercises 2. Quiz **Target: 60%**

Note: All syllabi without the UCIO control identifier and stamp for circulation are uncontrolled.

23	**MIDTERM EXAMINATION** **Target: 60%**						
24-35	**CO2 & CO3** LO5: Use logic to communicate more effectively and construct valid arguments. LO6: Analyze information and their relationship. LO11: Determine the validity of arguments. LO7: Determine valid conclusions based on arguments/assumptions.	**Chapter 5: Logic** A. Logic Statement and Quantifiers B. Truth Tables and Tautologies C. Conditional, Biconditional and Related Statements D. Symbolic Arguments Arguments and Euler Diagrams	1.Lecture through PowerPoint presentation. 2. Send learning materials in our LMS/Group Chat 3. Ask them to watch a recorded video lecture via YouTube channel	Workbook, reference books	Quizzes and problem sets	1. Questionnaires 2. Use of scoring rubrics	1. Activities/ Exercises 2. Quiz **Target: 60%**
	CO2 & CO3 LO8: Define a mathematical system LO9: Discuss examples of algebraic and abstract mathematical systems	**Chapter 6: The Mathematical Systems** A. Algebraic Mathematical Systems B. Abstract Mathematical Systems C. Modular Arithmetic D. Application of Modular Arithmetic E. Introduction to Group Theory	1.Lecture through PowerPoint presentation. 2. Send learning materials in our LMS/Group Chat 3. Ask them to watch a recorded video lecture via YouTube channel	Workbook, reference books	Quizzes and problem sets	1. Questionnaires 2. Use of scoring rubrics	1. Activities/ Exercises 2. Quiz **Target: 60%**

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	CO2 & CO3 LO10: Construct different types of graphs LO11: Distinguish the difference between Eulerian and Hamiltonian graphs LO12: Apply Euler and Hamiltonians paths to solve problems LO13: Explore the application of graphs LO14: Solve problems using graphs	**Chapter 7: The Mathematics of Graphs** A. Graphs and Euler Circuits B. Weighted Graphs C. Planarity and Euler's Formula Graph Coloring	1.Lecture through PowerPoint presentation. 2. Send learning materials in our LMS/Group Chat 3. Ask them to watch a recorded video lecture via YouTube channel	Workbook, reference books	Quizzes and problem sets	3. Questionnaires Use of scoring rubrics	1. Activities/ Exercises 2. Quiz **Target: 60%**
	CO2 & CO3 LO15: Describe coding schemes that are used to assign identification numbers LO16: Check digits and detect code errors LO17: Discuss the process of disguising data LO18: Analyze encrypted data	**Chapter 8: Codes** A. Coding B. Cryptography	1.Lecture through PowerPoint presentation. 2. Send learning materials in our LMS/Group Chat 3. Ask them to watch a recorded video lecture via	Workbook, reference books	Quizzes and problem sets	4. Questionnaires Use of scoring rubrics	1. Activities/ Exercises 2. Quiz **Target: 60%**

			YouTube channel				
36	**FINAL EXAMINATION** **Target: 60%**						

MAJOR COURSE OUTPUT/REQUIREMENTS	**GRADING SYSTEM**
1. Portfolio 2. Major Examination 3. Quizzes 4. Exercises 5. Assignment 6. Activities **Target: 60% (No. of students who will pass)**	Major Examination 40% Quizzes 30% Worksheets/Problem Solving 15% Assignment and Project 15% Total: **100%**

REFERENCES/OPEN EDUCATIONAL RESOURCES (OERs)		**Other Suggested Readings**	
1. Mathematics in the Modern World by R. Aufman, et al (Cengage Philippine Edition) 2. Mathematics in the Modern World by Ethel Cecile Baltazar, et al 3. Mathematics in the Modern World by Winston S. Sirug 4. Mathematics in the Modern World by Faustino E. Oguan Jr., et al 5. Mathematics in the Modern World by Mary Joy J. Rodriguez	**Ot** htt htt	6. Mathematics in the Modern World by R. Aufman, et al (Cengage Philippine Edition) 7. Mathematics in the Modern World by Ethel Cecile Baltazar, et al 8. Mathematics in the Modern World by Winston S. Sirug 9. Mathematics in the Modern World by Faustino E. Oguan Jr., et al 10. Mathematics in the Modern World by Mary Joy J. Rodriguez	**Ot** htt htt

CLASS POLICIES

CONSULTATION SCHEDULE						
Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Time						

CONTINUOUS QUALITY IMPROVEMENT
To ensure the continuous enhancement of the curriculum, the following forms/tools are available from the University Curriculum and Instruction Office (UCIO) for course review and monitoring: 1. RTU-OVPASA-UCIO-F018 (Syllabus Development and Revision Request Form) 2. RTU-OVPASA-UCIO-F019 (Course Syllabus Acknowledgement Form)

3. RTU-OVPASA-UCIO-F035 (Program Outcomes Assessment and Evaluation Matrix)
4. RTU-OVPASA-UCIO-F036 (Stakeholder Consultation Survey Form)

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